

Numerical Solutions to Problems in Electrochemistry



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Abstract

This is a project on PINNs that do not seem to work very well.

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Chapter 1

Introduction

Electrochemistry is a branch of chemistry that is concerned with the relationship between electrical applied potential and chemical change. These reactions arise at the interface between an electronic conductor and an ionic conductor, where charge transfer drives oxidation and reduction of chemical species. The study of these reactions has applications in biochemical sensing, energy storage, technology and many others that underpin much of modern biology and nanotechnology. In particular, electroanalytical techniques have enabled the detection of trace analytes in biological fluids, the development of rechargeable batteries and fuel cells, and the fabrication of nanoscale electronic devices, underlining the broad relevance of electrochemical science across disciplines.

Voltammetry is a common technique to analyse an electroactive species in solution. Voltammetry measures the current that is induced into an electrochemical circuit as the potential of the working electrode is varied with respect to the reference electrode. The current response arises from Faradaic processes, in which electrons are transferred between the electrode surface and the electroactive species, and its magnitude reflects both the kinetics of this charge transfer and the rate at which species are transported to the electrode. Voltammetric techniques are distinguished by how the applied potential is varied over time. In cyclic voltammetry, the potential is swept linearly between two limits and then reversed, producing a characteristic current–potential curve from which thermodynamic and kinetic parameters may be inferred. Other waveforms, such as the sinusoidal perturbations of electrochemical impedance spectroscopy or the staircase profiles of square-wave voltammetry, offer complementary sensitivity and resolution.

Voltammetry has been extensively studied mathematically and numerically. The electron transfer kinetics at the electrode surface are described by the Butler–Volmer

model, which relates the rate of the electrode reaction to the applied overpotential through exponential anodic and cathodic contributions, or alternatively by the Marcus–Hush framework, which grounds the kinetics in a quantum-mechanical treatment of solvent reorganisation. At equilibrium, the Nernst equation dictates the ratio of oxidised to reduced species concentrations at the electrode surface as a function of the applied potential. The transport of electroactive species through the solution is governed by Fick’s laws of diffusion, which, combined with the electrode boundary conditions described above, yield a partial differential equation (PDE) that fully characterises the evolution of the concentration profile in the cell. The solution of this system—the so-called forward problem—predicts the current response given a known set of physical parameters, and has been extensively studied both analytically and numerically [2].

We are interested in the inverse of this process: given the measured current response from a voltammetric experiment, we wish to infer the underlying physical parameters that characterise the electroactive species in solution. These parameters may include the formal potential, diffusion coefficients, and electron transfer rate constants, knowledge of which provides quantitative insight into the thermodynamics and kinetics of the redox system under study. This constitutes the inverse problem, and its solution is considerably more challenging than the forward problem. The mapping from parameters to current is nonlinear and the measurements are corrupted by noise, meaning that naive inversion is typically ill-posed and that uncertainty in the inferred parameters must be carefully quantified.

Many different techniques have been employed to solve this inverse problem. Deterministic optimisation methods, such as least-squares fitting, provide point estimates of the parameters but offer no principled account of uncertainty. Bayesian inference addresses this limitation by treating the parameters as random variables and seeking their posterior distribution conditioned on the observed data. Markov chain Monte Carlo (MCMC) methods have emerged as the dominant computational approach to posterior sampling in electrochemistry, enabling rigorous uncertainty quantification without requiring analytical tractability of the posterior [3]. However, standard MCMC algorithms such as the random-walk Metropolis–Hastings sampler can mix poorly in high-dimensional or geometrically complex parameter spaces, and the requirement to repeatedly solve the forward PDE makes the burn-in phase computationally expensive.

With the advent of modern automatic differentiation (AD) software, it is now possible to efficiently compute exact derivatives of arbitrary numerical programs,

including complete PDE solvers, with respect to their inputs [1]. This capability opens new avenues for gradient-based inference in electrochemistry. In particular, the gradient of the simulated current with respect to the model parameters can be obtained at a cost comparable to a single forward solve, enabling the use of gradient-informed algorithms both during the burn-in phase and throughout the sampling procedure. During burn-in, gradient information can be exploited to drive the chain rapidly toward regions of high posterior probability, dramatically reducing the number of forward solves required before convergence. Once in the posterior, gradient-based samplers such as Hamiltonian Monte Carlo (HMC) use the gradient to make more informed proposals, resulting in higher acceptance rates and faster exploration of the posterior landscape compared with gradient-free alternatives.

This dissertation demonstrates the applicability of gradient-based methodologies to solve the inverse problem in electrochemical experiments. We develop a differentiable PDE solver for common voltammetric waveforms, validate it against known analytical results, and integrate it with gradient-informed MCMC algorithms to perform Bayesian inference of electrochemical parameters. We show that the use of automatic differentiation yields substantial improvements in computational efficiency over gradient-free methods, and we illustrate the approach on both synthetic and experimental data.

Chapter 2

Electrochemical Systems

2.1 Voltammetry

In a typical cyclic voltammetry experiment the applied potential is swept linearly with time from some starting potential E_i where the species A is stable to some other, more negative potential, E_v , at which electron transfer occurs rapidly to form species B. The potential is then swept back to E_i causing electron transfer in the opposite direction and the reformation of A.

Throughout this process the current, I , is recorded. Plotting the current induces against the applied potential gives us a voltammogram.

This technique is useful as the resulting peak shaped signal provides a signature of the features of the reduction and oxidation processes. Analysis of the position and shape of the peaks can give important information about the nature of the electrochemical process taking place and about the chemical species themselves.

The scan rate, ν , (Vs^{-1}), is the constant rate at which the voltage sweeps from the initial potential to the vertex potential and back again.

$$\nu = \frac{dE}{dt} \tag{2.1}$$

As seen in Figure 2.1

2.2 Mathematical Description

We can represent a typical reduction process at the electrode as



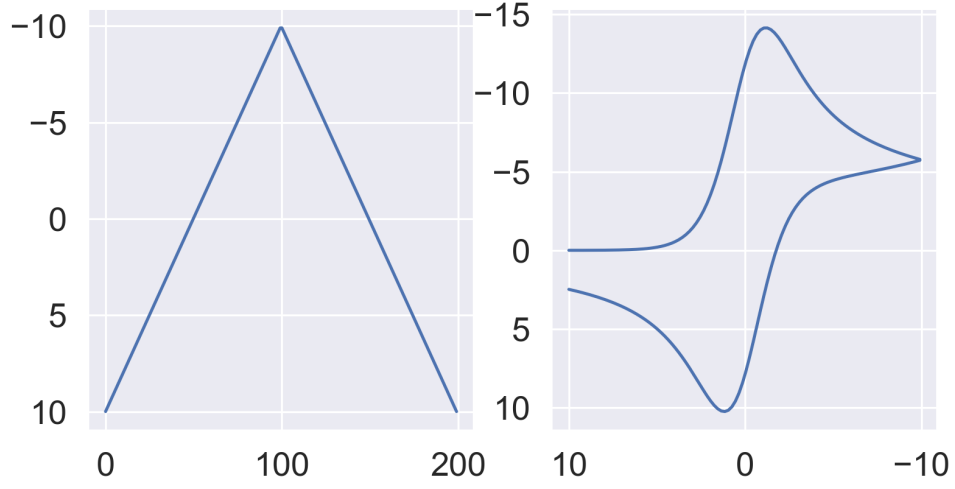


Figure 2.1: Left applied potentials and right is the voltammogram

2.3 Mass Transport

For a macroband electrode:

$$\frac{\partial c_j}{\partial t} = D_j \frac{\partial^2 c_j}{\partial x^2} \quad (2.3)$$

Where D_j is the diffusion coefficient associated with the species j . In the case of Equation 2.2, we obtain a mass transport equation for each of species A and B:

$$\begin{cases} \frac{\partial c_A}{\partial t} = D_A \frac{\partial^2 c_A}{\partial x^2} \\ \frac{\partial c_B}{\partial t} = D_B \frac{\partial^2 c_B}{\partial x^2} \end{cases}$$

2.4 Boundary Conditions

Initially, we will assume we have only the species A in solution which is equal to the bulk concentration, c_A^* and that we have none of species B.

2.5 Equal Diffusion Coefficients

Fick's first law states that the diffusive flux, j_{x_1} , through a plane normal to the x -axis at the point x_1 is equal to the product of the diffusion coefficient and the concentration gradient at that point.

$$j_{x_1} = -D \left(\frac{\partial c}{\partial x} \right)_{x=x_1} \quad (2.4)$$

Where the minus sign indicates the flux is down the local concentration gradient. We assume that initially the experiment contains only species A at a uniform concentration, the bulk concentration indicated by c_A^* . By conservation of mass, the amount of A lost at the electrode surface equals the amount of B gained:

$$\left(-D_A \frac{\partial c_A}{\partial x} \right)_{x=0} = \left(D_B \frac{\partial c_B}{\partial x} \right)_{x=0} \quad (2.5)$$

If we assume equal diffusion coefficients $D = D_A = D_B$. Then the problem simplifies to:

$$\left(-\frac{\partial c_A}{\partial x} \right)_{x=0} = \left(\frac{\partial c_B}{\partial x} \right)_{x=0} \quad (2.6)$$

Therefore it is necessarily true that: $c_A^* = c_A + c_B$. So we can only model the concentration of A and automatically recover the concentration of B. We will now be dropping the subscript of the species and assume we are always referring to A.

2.6 Initial Conditions

As already mentioned, we are assuming that initially there is only A present in the solution. Thus the initial conditions are:

$$c(x, t = 0) = c^* \quad (2.7)$$

2.7 Boundary Conditions

In the spacial dimension we use Einstein's work on Brownian motion. $0 \leq x \leq 6\sqrt{Dt_{\max}}$.

The first boundary condition we say that there is no change in the concentration when we are sufficiently far away from the surface of the working electrode:

$$c(x = x_{\max}, t) = c^* \quad (2.8)$$

At the surface electrode we have the Butler-Volmer equation:

$$\begin{aligned} \left(-\frac{\partial c}{\partial x}\right)_{x=0} = k_0 \left[c_A \exp\left(-\alpha \frac{F}{RT} (E_{\text{eff}}(t) - E_0)\right) \right. \\ \left. - (1 - c) \exp\left((1 - \alpha) \frac{F}{RT} (E_{\text{eff}}(t) - E_0)\right) \right] \end{aligned} \quad (2.9)$$

Chapter 3

Numerical Methods

3.1 Finite Difference Methods

3.2 Automatic Differentiation

Chapter 4

Markov chain Monte Carlo and the Inverse Problem

4.1 Metropolis-Hastings

4.2 Random Walk Metropolis-Hastings

4.3 Hamiltonian Monte Carlo

Chapter 5

Electron Transfer Reaction

Chapter 6

Heterogeneous Reaction

Chapter 7

Adsorption Reaction

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